



A mixed logit model for predicting exit choice during building evacuations



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ABSTRACT

Knowledge on human behaviour in emergency is crucial to increase the safety of buildings and transportation systems. Decision making during evacuations implies different choices, of which one of the most important concerns is the escape route. The choice of a route may involve local decisions on alternative exits from an enclosed environment. This study investigates the effect of environmental (presence of smoke, emergency lighting and distance of exit) and social factors (interaction with evacuees close to the exits and with those near the decision-maker) on local exit choice. This goal is pursued using an online stated preference survey carried out making use of non-immersive virtual reality. A sample of 1503 participants is obtained and a mixed logit model is calibrated using these data. The model shows that the presence of smoke, emergency lighting, distance of exit, number of evacuees near the exits and the decision-maker and flow of evacuees through the exits significantly affect local exit choice. Moreover, the model indicates that decision making is affected by a high degree of behavioural uncertainty. Our findings support the improvement of evacuation models and the accuracy of their results, which can assist in designing and managing building and transportation systems. The main aim of this study is to enrich the understanding of how local exit choices are made and how behavioural uncertainty affects these choices.

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1. Introduction

Reducing the number of fatalities and injuries during evacuations from buildings and transportation systems is the main aim of fire safety engineering. This goal can be achieved by designing evacuation systems and procedures so that the time needed by evacuees to escape safely (required safe egress time) is shorter than the time from ignition to the moment when the conditions of the given environment become untenable (Available Safe Egress Time). To date, several evacuation models have been developed to estimate the required safe egress time simulating human behaviour in fire (Gwynne et al., 1999; Kuligowski et al., 2010).

The evacuees' behaviour can be seen as the result of a hierarchical decision-making process entailing three stages: (1) *strategic* (choice to go towards a safe place); (2) *tactical* (choice of routes and exits) and (3) *operational* (short-range choices

Abbreviations: IBU, intrinsic behavioural uncertainty; PPBU, perceptions and preferences behavioural uncertainty; ED, efficient design; MLM, mixed logit model; MNL, multinomial logit; RP, revealed preference; RUM, random utility model; RUT, random utility theory; ST, stated preference.

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concerning the interaction with obstacles and other evacuees) (Lovreglio et al., 2014, 2015a, 2015b). The literature argues that escape route (i.e. tactical choices) can determine the effectiveness of the evacuation process in a crucial way (Ronchi, 2012; Nilsson, 2009; Lovreglio, 2014; Fridolf et al., 2013; Huang and Guo, 2008; Lo et al., 2006; Zheng et al., 2009; Gwynne et al., 2001; Heliövaara et al., 2012). From a modelling point of view, the decision concerning the route to a safe place entails global and local choice (Ronchi and Nilsson, 2016). In fact, evacuees try to select the final goal(s) of their 'evacuation journey' through the global exit choice and then they try to achieve the selected goal making local exit choices. For example, the final/global goal could be to reach a specific exit of a building, whereas the local exit choices are made to pursue the final/global goal. However, although evacuees can be familiar with the building, it is not always realistic to assume that they have a complete knowledge of the global escape route. There could be situations in which the global evacuation route may be the consequence of local choices, as different local exits from the same environment may lead to very different global escape routes (Gwynne et al., 2001; Wagoum et al., 2011).

Several environmental, social and personal factors can affect the global and local exit choice during emergencies (Lovreglio et al., 2014). The most influential environmental factors are (a) distance from the exits, (b) fire conditions (e.g. visibility of an exit; presence of smoke or flames close to an exit) and (c) emergency lighting (Nilsson, 2009; Ronchi et al., 2012, 2015; Gwynne et al., 2000). Different types of social influences can also affect exit choice leading to different behaviours: *herding behaviour*, *leader–follower behaviour*, *cooperative behaviour* and *competitive/selfish behaviour* (Lovreglio et al., 2014, 2016). These social behaviours have been interpreted qualitatively using several theories: (1) the *role–rule theory*, explaining the behaviour based on the behavioural rules of the evacuees, which depend on their daily roles (e.g. staff of a transportation system may react differently from the users) (Canter et al., 1980; Tong and Canter, 1985); (2) the *affiliative theory*, focusing on the decision-maker's attitude to follow familiar evacuees (Sime, 1985); (3) the *social influence theory*, arguing that other evacuees are a source of information (*informational social influence*) and the decision-maker aims to conform his/her choice to that of other evacuees, to avoid their negative judgment (*normative social influence*) (Nilsson and Johansson, 2009) and (4) the *social proof theory*, according to which a decision is considered correct by the decision-maker, because other evacuees have already taken it (Cialdini, 1993). Besides the environmental and social factors, personal factors can affect exit choice. The most influential personal factor is the familiarity of the decision-maker with an exit (*affiliation behaviour*) (Sime, 1985; Shields and Boyce, 2000; Nilsson et al., 2008; Toyama et al., 2006; Fahy et al., 2012; Proulx, 1993; Wagoum, 2012). Furthermore, physical ability (depending on age or health), handedness and socio-psychological characteristics (for instance, direct or indirect risk perception, cultural background or training and past experience) can affect the exit choice (Lovreglio et al., 2014; Ronchi et al., 2012; Wagoum, 2012; Nilsson et al., 2009; Veeraswamy et al., 2011).

A key issue in modelling and designing for evacuations is generally a lack of consideration of the stochastic nature of human behaviour (Ronchi et al., 2013; Lovreglio et al., 2014). The behavioural uncertainty is due to two sources of randomness: the 'intrinsic behavioural uncertainty' (IBU) and the 'perceptions and preferences behavioural uncertainty' (PPBU). IBU is based on the facts that (a) the choices taken by different decision-makers perceiving a situation in the same way may be different and (b) the same decision-makers could choose different exits when they face the same situation at different times. PPBU is related to different decision-makers' perceptions (i.e. different decision-makers can have different quantitative estimates of the same factor) and preferences (i.e. a certain factor may have different importance to different evacuees) concerning the factors affecting the choice. Therefore, behavioural uncertainty represents a key feature that needs to be included in evacuation models. In order to enrich the understanding of how behavioural uncertainty may affect the decision-making process, further studies are necessary.

Here, a case study of local exit choice during an evacuation from an enclosed environment with two exits is presented. This study investigates the effect of both environmental and social factors on exit choice, including the presence of other evacuees, fire conditions, emergency lighting and distance from the exit. The study is based on an online stated preference survey using non-immersive virtual reality scenarios. Responses from 1503 respondents have been collected from all over the world. Choices are modelled using the random utility theory (RUT), which assumes that the decision-maker chooses the alternative yielding the maximum utility and that this utility is not completely known to the modeller, so it has to be considered partially stochastic (Ortuzar and Willumsen, 2011; Train, 2009; Hensher et al., 2005). Therefore, the main aim of this study is to provide new experimental data, which allow expanding and enriching the current understanding of local exit choice in emergencies, and to verify the importance of the behavioural uncertainty in local exit choice.

The study begins with an introduction of existing approaches to model exit choice, supporting the use of the RUT and discussing the underpinning assumptions (Section 2). Section 3 introduces the methodological steps used in the case study. The survey is presented in Section 4, which provides details on the design and administration of the questionnaire and the obtained sample. The proposed exit choice model is introduced in Section 5 and discussed in Section 6, including a sensitivity analysis of the model. The conclusions in Section 7 discuss the practical implications of our study and future works.

2. Methodological issues

Different approaches have been adopted to model exit choice (Kuligowski et al., 2010). Section 2.1 provides a general overview and supports the use of the RUT in this study. The modelling assumptions underpinning the RUT are introduced in Section 2.2, where models using the RUT are reviewed to justify the need for new model specifications/calibration.

2.1. Approaches to exit choice modelling

Three categories of exit choice rules are considered in existing evacuation models:

- (a) Agents (i.e. simulated evacuees) head towards exits predefined by the modeller;
- (b) Agents choose the closest exit and
- (c) Agents choose the exit considering environmental, social and personal factors (Kuligowski et al., 2010; Gwynne et al., 2000; Wagoum, 2012; Schneider and Könnecke, 2010).

The first approach is clearly limited, because it does not consider any evolution of the evacuating scenarios and the choice is a user input rather than an output of the model (Gwynne et al., 2015). In the second one (distance-based model), the choice is context-dependent but static and based only on the building structure. It does not allow for dynamic adjustments to avoid congestion (Schneider and Könnecke, 2010). The third category of models entails that each agent evaluates the features of the simulated environment and takes decisions based on the perceived information. In these models, the chosen exit can change during the evacuation process if the evacuation conditions change and a range of factors can be considered (e.g. presence of smoke, visibility, familiarity with an exit). The simplest and most common model of the third category is the time-based one, in which the agents choose the exit with the least evacuation time.

The modelling approaches to exit choice can be classified into deterministic and stochastic (Lovreglio et al., 2014). Deterministic approaches have been derived from different decision theories, such as the game theory (Lo et al., 2006; Ehtamo et al., 2010; Mesmer and Bloebaum, 2014) or the utility maximisation theory (Wagoum, 2012; Ehtamo et al., 2009). Deterministic models can represent only average behaviours. By contrast, stochastic models take behavioural uncertainty into account. Several stochastic approaches have been used for exit choice. For instance, Zhang et al. (2013) introduced an exit choice model, in which the ‘base probability’ of using an exit is defined by the modeller. However, these pre-defined probabilities may change depending on the previous use of the exit and the fire condition of the next compartment connected to the exits. This approach requires prior knowledge of usage probabilities, which can be difficult to obtain. This issue can be overcome by random utility models (RUMs), as they do not require any pre-defined probability.

There are two main reasons for the adoption of the RUT as the modelling framework in this study. On the theoretical side, both IBU and PPBU can be taken into account. On the implementation side, well-established techniques exist, to calibrate RUMs from stated preference (SP) or revealed preference (RP) surveys (Ortuzar and Willumsen, 2011; Greene, 2011). The RUMs implemented to analyse the results of our survey are discussed in the following section.

2.2. RUM framework

In the RUT framework, the decision-maker assigns to each available option a utility, which depends on the relevant attributes of the option itself. The option with the highest utility is more likely to be chosen. To consider the behavioural uncertainty, it is assumed that the utility U_{iq} of the i alternative for the q decision-maker consists of two terms:

$$U_{iq} = V_{iq} + \varepsilon_{iq} \quad (1)$$

where V_{iq} is a deterministic component, whereas ε_{iq} is a random one (i.e. random residual) (Ortuzar and Willumsen, 2011). In this study, a linear specification is used for the deterministic part:

$$V_{iq} = \beta_{ij} X_{ij} \quad (2)$$

where X_{ij} are the known values of the j factors perceived by the q decision-maker affecting the choice for the i alternative, whereas β_{ij} are weights representing the decision-maker preferences related to j factors and are to be estimated. The functional form of the probability of choosing each option depends on the hypothesis of the distribution of the random residual. The widely used multinomial logit models (MNL):

$$P_{iq} = \frac{e^{V_{iq}}}{\sum_k e^{V_{ik}}} \quad (3)$$

derives from assuming that random residuals have Gumbel distributions with mean 0 and variance $\pi^2/6$ and these are independent and homoscedastic (Ortuzar and Willumsen, 2011). The standard logit approach considers IBU by introducing the random residual term and it is simple to implement. The residual term includes also the modeller's error (i.e. the lack of knowledge of the relevant factors affecting the decision) (Ortuzar and Willumsen, 2011; Cascetta, 2009). The MNL assumes that preferences/tastes are constant across evacuees and deterministic; therefore, PPBU is not taken into account. PPBU is instead considered in random parameter models, such as mixed logit models (MLMs) (Train, 2009). The MLM approach assumes that β_{ij} are randomly distributed because of decision-makers' different tastes and perceptions of a single factor. Therefore, the probability of choosing the i alternative by q decision-maker is

$$\tilde{P}_{iq} = \int \frac{e^{V_{iq}}}{\sum_k e^{V_{ik}}} f(\beta_{ij} | \alpha_{ijz}) d\beta_{ij} \quad (4)$$

where f is the probability density function of the β_{ij} coefficients and α_{ijz} is the z parameters of f (Train, 2009; Greene and Hensher, 2010). In general, the MLMs have no closed solution. However, the probabilities can be estimated by using Monte Carlo techniques (Train, 2009; Greene and Hensher, 2010). Let β_z denote vectors of β_{ij} coefficients drawn from f . An estimation of the probability that the q decision-maker selects the i alternative can be calculated by randomly drawing R vectors β_z , calculating the corresponding values of P_{iq} and then averaging according to the following equation:

$$\bar{P}_{iq} = \frac{1}{R} \sum_z^R P_{iq}(\beta_z) \quad (5)$$

$\bar{P}_{iq}$ can be then used to estimate α_{ijz} by maximising the likelihood function. The likelihood for Q decision-makers can be written as

$$L = \prod_{q=1}^Q \sum_i^{I_q} y_{iq} \cdot \bar{P}_{iq} \quad (6)$$

where y_{iq} is equal to 1 if the q decision-maker ($q = 1, \dots, Q$) selects the i alternative ($i = 1, \dots, I_q$), otherwise it is 0. Numerous techniques are available in the literature to solve the likelihood maximisation problem (Ortuzar and Willumsen, 2011; Hensher et al., 2005; Greene, 2011).

RUMs have been already used for modelling exit choice. Huang and Guo (2008) proposed an MLM which predicted the probability of choosing an exit as a function of the distance associated to each available exit. The multinomial logit formulation was adopted also by Guo and Huang (2010), whose model considered both free flow (related to the exit distance) and congestion (number of evacuees approaching the exit) disutilities, and the exit width (that is an indirect measure of the flow through an exit). Different from the studies described below, neither model was calibrated using experimental data.

Duives and Mahmassani (2012) investigated the effects of exit distance, angular deviation, total number of evacuees, number of evacuees near the exit and decision-maker handedness. A binary logit model was estimated using data collected through an online SP survey including 16 hypothetical scenarios. The sample included 117 participants from the Netherlands and the United States. The results showed that exit distance, angular deviation and total number of evacuees significantly affect exit choice.

Lovreglio et al. (2014) studied the effect of the number of evacuees close to each exit and to the decision-maker, and the position of the decision-maker using a proxy measure of distance (i.e. close to an exit and far from an exit). A mixed binary logit was estimated using the choices of 191 Italian respondents, who were presented with 12 hypothetical scenarios in an online survey. All the environmental factors were found significant. The survey showed also that age, height and education affected the perception of the distance from the exit and the impact of the evacuees near the decision-maker. Finally, the study proved the heterogeneity among respondents of the perception and preference (PPBU) concerning the number of evacuees close to each exit and the distance from the exit.

Haghani et al. (2014) investigated the effect of exit distance, density around each exit, flow towards each exit and exit visibility using data collected from face-to-face interviews with 53 Australian respondents. A hybrid survey technique combining SP and RP (Train and Wilson, 2008) was used to analyse the choices made in one real and 14 hypothetical scenarios. The authors estimated an MLM, proving that all the factors, with exception of the exit flow, were statistically significant. Also in this survey PPBU was observed.

Lovreglio et al. (2014) focused on herding behaviour, that is, the attitude of respondents to follow the decision taken by the majority of evacuees. Applying an MLM, the authors showed that herding behaviour was affected by both environmental (number of evacuees near the two exits) and personal (gender, weight and occupation) factors. Finally, Lovreglio et al. (2016) refined the model in Lovreglio et al. (2014) by introducing different herding classes. The study showed the existence of heterogeneity in the herding attitude.

It is noted that each of these studies considered a subset of the potential influencing factors at a time and besides, the calibration was based on surveys in which the scenarios were represented in a simple way. This study improved the current knowledge by considering more factors than the existing studies, and by using virtual reality to better represent scenarios. The inclusion of more factors in the study allows a better estimation of the relative effect of these factors. The use of virtual reality significantly improved the realism of the experience of the respondents during the survey.

3. Methodological steps

RUMs can be calibrated using SPs or RPs (Ortuzar and Willumsen, 2011). In the SP approach, hypothetical scenarios are proposed to the participants in the study. Researchers can control for the variables deemed relevant in SP experiments and data collection is relatively quick and cheap (costs can increase though when face-to-face interviews are used for surveying). However, the data collected by SP approach may be biased because the interviewees do not face a real context (i.e. the results may have low ecological validity) (Kinateder et al., 2014). The RP approach does not have this shortcoming, but data from actual evacuations are often difficult to obtain. Moreover, even when real data (in the form of videos) are available, there are two severe limitations. First, researchers have no control over the sample and the variables affecting the choice. Second, the emotional state and the mental processes of the evacuees cannot be analysed directly but only induced from the

behaviour in emergencies. Interviews with people involved in the evacuation may help overcome the latter limitation, but interviews can hardly be related to the data extracted from the videos (Lovreglio et al., 2015). An SP experiment based on virtual reality is considered suitable for this study. The experimental control of this type of survey allows researchers to investigate the impact of several independent variables by collecting data suitable for the estimation of 'good models' (Ortuzar and Willumsen, 2011). The scenarios presented in SP experiments can be designed to have (a) sufficient variability of the independent variables and (b) low collinearity of these variables (Louviere et al., 2000).

In this study, we deal with social and environmental factors only, whereas the study of the effect of personal attributes on exit choice (except gender, see Section 5) is left to future work. Personal factors can indeed affect exit choice, but their influence is deemed not relevant here, because our aim is to investigate the behavioural uncertainty related to social and environmental factors.

Our study includes three steps: (1) background and pilot study, (2) final survey design and (3) data collection and modelling (Fig. 1). In the first step, the variables which may affect exit choice were identified through literature review and analysing the interviews done during a previous study on exit choice (Lovreglio et al., 2014). Then, an online pilot survey involving 88 participants was carried out to both improve the representation of the scenarios and collect information for the design of the final survey. Face-to-face semi-structured interviews with some of the respondents provided insights into the perception of the contexts of choice and the involved variables. In the second step, the information collected through the pilot survey was used to define the levels of the variables characterising the scenarios in the final survey, using the efficient design (ED) technique explained below. Moreover, the results of interviews were used to improve the videos so that respondents could have an accurate perception of the contexts of choice. In the last stage, data collection was performed through an online survey. The videos representing the choice scenarios could be easily shown to respondents using the Internet. In addition, online survey allowed collecting data from large and heterogeneous samples in short time and with very small costs. The responses were analysed using an MLM, taking into account both IBU and PPBU.

The scenarios used in SP surveys were defined by the levels assumed by the relevant factors. When the number of variables is large, the number of the scenarios generated by the combinations of all levels of all variables becomes easily intractable. To cope with this issue, we used the ED technique to select the scenarios to be included in our survey (Greene, 2011; Rose et al., 2008; Sándor and Wedel, 2001; Institute of Transport and Logistics Studies, 2007). This method is based on the minimisation of the so-called D-error, which is the determinant of the asymptotic variance–covariance matrix (i.e. the negative inverse of Hessian matrix of the log-likelihood function) to the power of $1/K$, where K is the number of parameters to

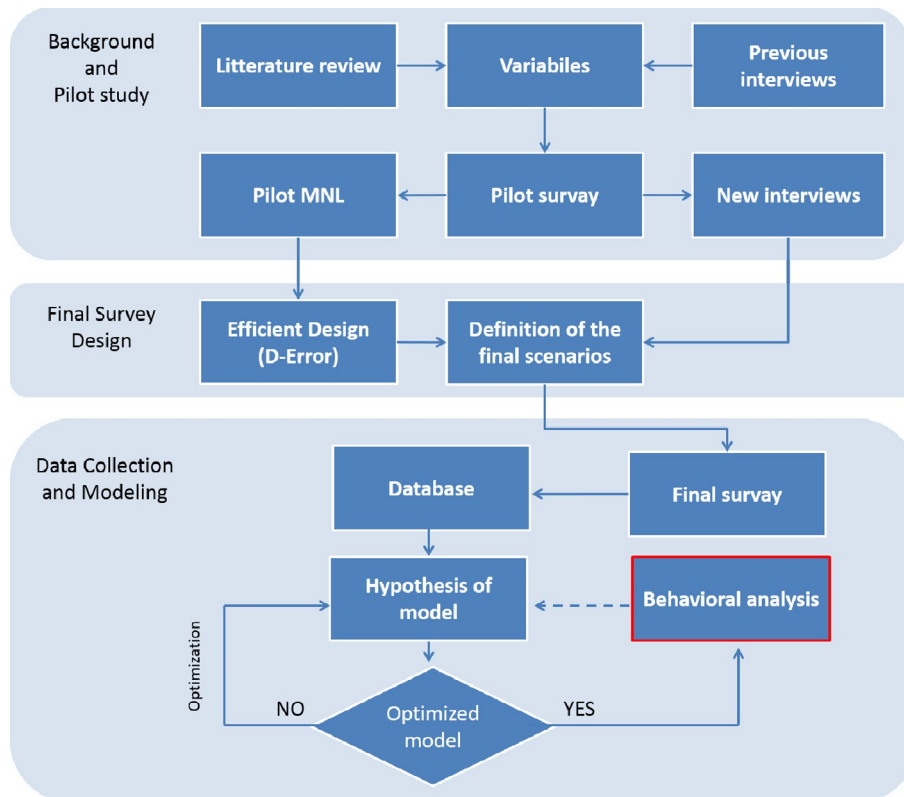


Fig. 1. Methodological scheme.

estimate. Therefore, D-error is related to the p values of the parameters to estimate, because p values are calculated using the variance matrix (i.e. the diagonal elements of variance–covariance matrix) (Greene, 2011). In order to implement ED, approximated values of the model parameters ('prior' values) are necessary before running the survey. Prior values can be found in the literature or, if not available as in our case, obtained from a pilot study. The pilot survey can be designed by means of ED, using educated guesses on the sign and the value of each parameter to estimate.

4. SP experiment

4.1. Contexts of choice

The context of choice used in this study included two exits, one on the left-hand side and other on the right-hand side of the decision-maker. These exits were set in an enclosed environment similar to a metro station with rectangular plant (size: 23×18 m) as shown in Fig. 2. The hypothetical scenarios were proposed using videos to make the context of choice more realistic. Moreover, videos allowed providing respondents with information deriving from the dynamic evolution of the evacuations: for instance, the capacity of an exit could be evaluated from the number of evacuees that flow through it in a fixed time. The videos were generated using Unity 3D (Personal Edition). The geometry of the metro station was built directly in Unity 3D, whereas evacuees' 3D bodies were downloaded from the Internet and their original file formats were converted using Blender. Pieces of code in C# were used to animate these virtual evacuees. In order to improve the realism, a fire alarm and voices were added (Video 1). During the experiment, decision-makers were supposed to be inside the environment and that videos were taken from their viewpoint.

It has been highlighted that exit choice can be affected by environmental (concerning the physical features of the choice context), social (related to the presence of other evacuees) and personal factors. Social influences and the descending behaviours are a particular focus of this study. The factors we considered are:

- **N**umber of evacuees **C**lose to the **E**xits (NCE);
- **F**low of evacuees through the exits (FL);
- **N**umber of evacuees **C**lose to the **D**ecision-**M**aker heading towards one of the exits (NCDM);
- **S**Moke near the exits (SM);
- **E**vacuation **L**ights above the exits (EL);
- **D**I**S**Tance of the decision-maker from the exits (DIST);

Fire conditions can affect the exit choice in two ways: the presence of the smoke near an exit can induce decision-makers to avoid that exit; the presence of smoke near an evacuee can affect his/her range of visual perception. In our videos, the presence of smoke does not affect the visibility of the exits (the respondents are able to see the exit and the simulated evacuees in every video). Therefore, we only study the effect of the presence of smoke near the exits, represented by a dummy variable (SM).

In the pilot survey, respondents were asked to point out the exit they would choose in 12 scenarios, defined by different levels of the variables as reported in Appendix A (see Tables A1 and A2). The pilot study involved 88 respondents corresponding to 1056 (88×12) observations. The starting values of the parameters for the design of the final survey with ED were estimated by calibrating an MNL model (Table 1). A total of 10 of the 88 respondents participated in a face-to-face semi-structured interview. Most of them stated that they could not perceive any difference between the flows though the two exits, which were actually different in some scenarios. This is confirmed by the MNL, where the parameter associated with FL is not significantly different from zero. In order to improve the perception of this variable, two very different levels were



Fig. 2. Frame from one of the videos.

Table 1
Pilot MNL.

	Coeff.	Std.Err.	t-ratio	P-value
NCE	−0.108	0.012	−9.108	0.000
FL	0.214	0.209	1.022	0.307
NCDM	−0.049	0.021	−2.318	0.020
SM	−0.985	0.123	−7.995	0.000
EL	0.175	0.101	1.736	0.082
DIST	−0.011	0.030	−0.362	0.718

chosen for the final survey (see Table 2). While in the pilot survey the different flows depend only on evacuees' speeds, in the final survey different flows were determined by both evacuees' speed and exit width. Even though the parameter associated with DIST is not significantly different from zero (see Table 1), almost all interviewees stated that they took into account the distance from the exits during the choice. Therefore, it was kept in the final survey.

The scenarios of the final survey were defined by different combinations of the variables and levels shown in Table 2. The subscript i represents the exit: L stands for the exit on the left-hand side and R for that on the right-hand side. NCE_i varies during the videos, because evacuees evacuate through the exits; the values shown in the table are those visible at the beginning of each video. The two dummy variables NEAR_E and DIR define respectively the position of the decision-maker and the direction of evacuees close to the decision-maker. NEAR_E = 0 if the decision-maker is closer to the right-hand-side exit and 1 otherwise. Similarly DIR = 0 if the evacuees near the decision-maker move towards the right-hand-side exit and 1 otherwise (Fig. 3).

Given the variable levels in Table 2, the number of possible scenarios (full factorial design) is $4^1 3^3 2^8 = 27,648$. The 12 scenarios listed in Table 3 were selected using ED. These scenarios were divided into two blocks of six and each respondent was presented with one of the two blocks. This allowed reducing the number of scenarios for each respondent and so to prevent respondents' fatigue, a problem indicated by some of the interviewees during the pilot survey, where participants were asked to state their decision in 12 cases. Moreover, the scenarios were presented randomly to avoid that the collected data could be biased by the order of the scenarios.

4.2. Survey

The questionnaire, in English and Italian, was disseminated through the Internet over a period of 2 months. The survey was advertised by e-mail lists and social networks (i.e. LinkedIn, Twitter and Facebook). This advertising strategy was used to collect as much data as possible from respondents coming from different parts of the world. The goal was to collect data from more than 450 respondents, which was the lower bound for the sample size suggested by the ED technique for this case study (Institute of Transport and Logistics Studies, 2007).

The survey included three sections. The first contained an introduction and demographic questions. In the second section, the videos representing the contexts of choice were shown. The respondents were instructed to make a choice at the end of the playback. It explicitly stated that *'there is no right and wrong choice and we are only interested in understanding what you would do in the situation you are faced with in the video'*. This was essential because the aim of the survey was not to collect data about the 'most rational/optimal' behaviour, but the 'natural' response to the situation. At the end of each video, respondents were directed to a new web page to choose between the left- and right-hand-side exits. This page also included a countdown timer that gave the respondents 5 s to answer the question. The countdown timer was used to prevent that excessively long reflection may lead to choices different from those in emergencies. In fact, respondents were allowed to state their choice also after the time runs out to reduce non-response, but they were not made aware of this aspect to keep the level of alertness high. Finally, in the third section, at the end of the videos, the respondents were asked questions about the level of realism of the proposed scenarios and their level of anxiety during the experiment.

Table 2
Levels for each variable.

Variable	Description	Levels		
NCE_i^a (pers)	Number of evacuees Close to the Exits	24	30	40
FL_i (pers/s)	Flow of evacuees through the exits	0.6	1.2	
NCDM (pers)	Number of evacuees Close to the Decision-Maker	0	5	10
SM_i	Smoke near the exits	0	1	
EL_i	Evacuation Lights above the exits	0	1	
DIST (m)	Distance of the decision-maker from the exits	10	12	14 16
NEAR_E	Dummy variable equal to 0 if the decision-maker is closer to the right-hand exit, 1 otherwise	0	1	
DIR	Dummy variable equal to 0 if the agents near the decision-maker move towards the right-hand exit, 1 otherwise	0	1	

^a The values refer to the beginning of the videos.

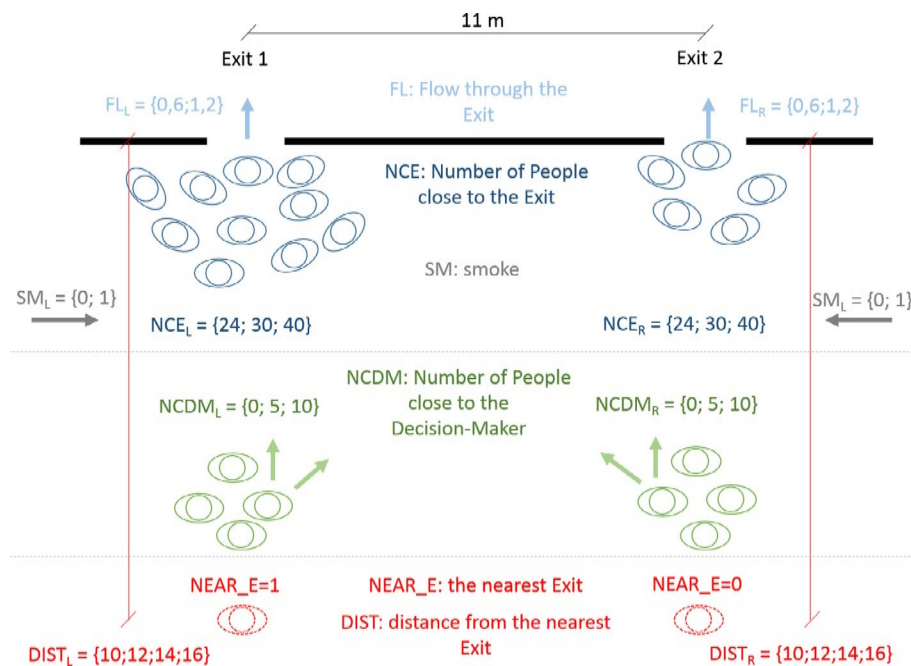


Fig. 3. Context of choice.

Table 3
Final scenarios.

Scenario	NCE _L	NCE _R	FL _L	FL _R	SM _L	SM _R	EL _L	EL _R	NCDM	DIR	DIST	NEAR_E	Block
1	24	30	1.2	0.6	0	1	1	1	10	1	14	0	1
2	40	40	0.6	0.6	1	0	1	0	5	0	16	1	1
3	30	30	0.6	1.2	0	0	0	1	10	0	10	0	1
4	24	40	1.2	0.6	1	0	0	1	0	1	10	1	1
5	40	24	1.2	1.2	0	1	0	0	0	1	12	1	1
6	30	40	0.6	1.2	1	1	1	0	0	0	16	0	1
7	40	40	0.6	0.6	0	0	0	0	5	1	10	0	2
8	40	24	0.6	0.6	0	1	1	1	5	0	12	1	2
9	24	30	0.6	1.2	1	1	0	1	10	1	16	1	2
10	30	30	1.2	1.2	1	0	1	1	0	1	14	0	2
11	30	24	1.2	0.6	1	1	0	0	10	0	12	0	2
12	24	24	1.2	1.2	0	0	1	0	5	0	14	1	2

4.3. Sample

Our sample is made up of 1503 respondents, corresponding to 9018 (1503 participants $\times$ six scenarios) observations. Female participants constitute 28.3% of the total study population. The mean age is 28.2 years, with standard deviation 11.4 years; 71% of the respondents are <30 years of age (Fig. 4). The majority of respondents are from Europe, mainly from Italy (22%) and the United Kingdom (11%) (Fig. 5).

The sample demographics are explained by the dissemination channels. In fact, the age distribution reflects the age distribution of social network users (Thelwall, 2008; Acquisti and Gross, 2006). However, the sample is large enough to take into account differences between male and female respondents as is discussed in Section 5.

4.4. Limitations

The data collected in this case study have some limitations. The major drawback is that the ecological validity of the results may be jeopardised by two issues:

- (1) The decision-makers were exposed to hypothetical situations with the awareness that they were involved in a trial, but not in a real emergency. This may imply that their answers are driven by their anticipation of what is the 'right' answer for the researcher, instead of reporting the 'natural' behaviour.

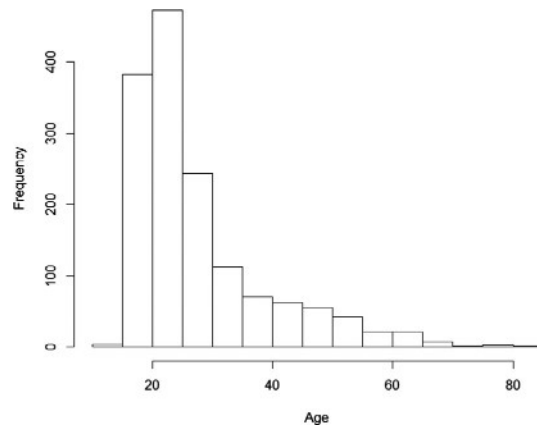


Fig. 4. Age distribution of respondents.

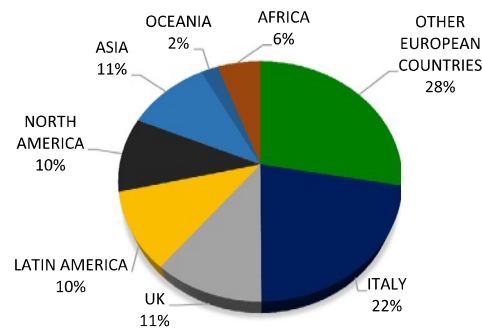


Fig. 5. Geographical distribution of respondents.

- (2) Despite the use of the countdown to increase respondents' stress through time pressure, it is impossible to induce with the physiological implications generated by a real emergency through an online survey based on a non-immersive virtual reality.

Previous studies have found that social behaviour in virtual reality simulations can be affected by behavioural realism and agency, that is 'whether an animated human is perceived as an agent or an avatar' (see Kinateder, 2012) according to the threshold model of social influence (Kinateder, 2012; Blascovich, 2002). These two factors are not taken into account in our survey. However, when the respondents are asked to rate the perceived general realism of the context of choice, they rate it 3.0 on average, in a scale from 1 – very realistic to 5 – not realistic at all.

The final limitation of the survey concerns the influence of the smoke. As explained in Section 4.1, we studied the smoke influence only by using a dummy variable indicating whether there was smoke near the two exits. However, it is evident that the visibility can play a key role in exit choice (Nilsson, 2009; Fridolf et al., 2013; Ronchi et al., 2012).

The effect of these limitations is discussed in the Conclusion section.

5. Estimated model

MNL and MLM are estimated using the data from the survey described in Section 4 (Table 4). Different from MNL, the parameters in MLM are normally distributed. Therefore, if the predictivity of the latter model is better than that of the former, it can be concluded that PPBU plays a key role in the exit choice process. The utility function of the exits includes all the factors described in Section 4. A constant is added to the utility function of the right-hand-side exit to check whether respondents are biased towards one of the two exits. NCE changes over the duration of the videos, because some evacuees leave the environment (with flow FL). In the model, we include the average values at the beginning and end of the simulation. Because there is a large difference between the number of female and male respondents, we have studied the interaction between a dummy variable defining the respondent gender (GEND = 1 if the respondent is female) and each environmental and social variable (V_i = NCE, FL, NCDM, SM, EX or DIST) to check if the gender statistically affects the choice. The model specification in Table 4 does not include the interaction terms, because they are not statistically different from zero ($p \gg 0.05$).

Table 4
Estimated MNL and MLM.

	MNL				MLM			
	Restricted log likelihood = −5419.025 Log likelihood function = −4357.728 RsqaAdj = 0.19440				Restricted log likelihood = −5419.025 Log likelihood function = −4284.297 RsqaAdj = 0.20798			
	Coeff.	Std.Err.	t-ratio	P-value	Coeff.	Std.Err.	t-ratio	P-value
NCE	−0.1161	0.0044	−26.4984	0.0000	−0.1713	0.0107	−16.0148	0.0000
FL	0.6092	0.0768	7.9348	0.0000	1.1455	0.1380	8.3008	0.0000
NCDM	−0.0771	0.0054	−14.1997	0.0000	−0.1041	0.0090	−11.5533	0.0000
SM	−0.7621	0.0512	−14.8936	0.0000	−1.0041	0.0852	−11.7836	0.0000
DIST	−0.0534	0.0066	−8.0633	0.0000	−0.0813	0.0114	−7.1273	0.0000
EL	0.8556	0.0397	21.5572	0.0000	1.2291	0.0853	14.4012	0.0000
CONST	0.0959	0.0271	3.5440	0.0004	0.0690	0.0364	1.8950	0.0581
NsNCE	/	/	/	/	0.0549	0.0105	5.2446	0.0000
NsFL	/	/	/	/	1.6450	0.2477	6.6411	0.0000
NsNCDM	/	/	/	/	0.0826	0.0171	4.8287	0.0000
NsSM	/	/	/	/	0.8860	0.1299	6.8194	0.0000
NsDIST	/	/	/	/	0.1972	0.0192	10.2540	0.0000
NsEL	/	/	/	/	1.1631	0.1228	9.4733	0.0000
NsCONST	/	/	/	/	0.4436	0.1013	4.3804	0.0000

In order to estimate MLM, a panel data approach is used to take into account the correlation between the answers of the same respondent who makes a choice in six different scenarios. A total of 300 Halton draws are used to simulate the random distribution of the parameters (Eq. (5)) (Train, 2009; Hensher et al., 2005).

In both models, all the factors are statistically significant (assuming a significance level of 0.05) except the constant of the MLM. The MLM shows that all the parameters are normally distributed. Fig. 6 shows the distribution of the different parameters.

The predictivity of the two models is studied through four indicators taken from the literature: the traditional log-likelihood and adjusted R-squared indicators, and two less common indexes (Train, 2009; Hensher et al., 2005; Greene, 2011; Cramer, 2003). The first of these latter two indicators is

$$E = \sum_{s=1}^S (P_s - F_s)^2, \quad (7)$$

where P_s is the predicted probability of choice of a scenario s and F_s is the corresponding observed frequency. Clearly $E = 0$ when the predictions perfectly match to the observation. The last considered indicator is

$$F = \frac{1}{N_{obs}} \sum_{i=1}^{N_{obs}} \delta_i, \quad \delta_i = 1 \text{ if } (P_{L,i} > P_{R,i} \wedge C_i = L) \vee (P_{R,i} > P_{L,i} \wedge C_i = R); \quad \delta_i = 0 \text{ otherwise}, \quad (8)$$

where N_{obs} is the total number of i observations; $P_{L,i}$ and $P_{R,i}$ are the predicted probabilities of choosing the left- and right-hand-side exits, respectively; C_i is the actual choice made by the decision-maker between the left- (L) and right-hand-side (R) exits.

Table 4 shows that the MLM outperforms the MNL model in the three first tests, whereas they have the same values of F because the exit with the highest probability of being chosen is the same in the two models. This result highlights that PPBU improves the predictivity of the local exit choice.

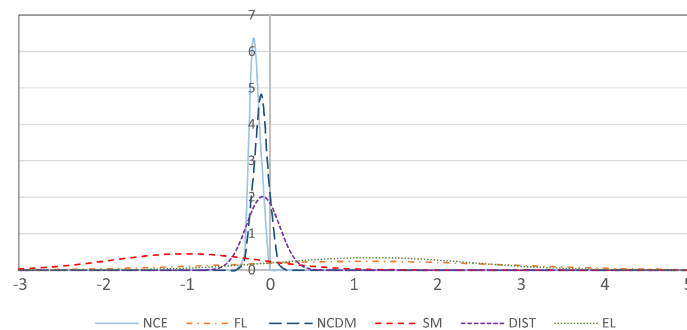


Fig. 6. Random distribution for (a) NCE and NCDM; (b) FL; (c) SM and DIST; (d) EL.

6. Discussion

The MLM proposed in this study allows predicting the probability of a decision-maker to choose an exit by considering six environmental and social factors. The analysis confirms the existence of PPBU. In fact, the perception of the factors and/or their relevance in the decision-makers is not constant among respondents, but the parameters (β_{ij}) associated to all the independent variables are normally distributed. It is important to note that different parameters distribution can be tested using the MLM approach. In the absence of evidence on the distributions in our case, we have selected the normal one, as it is commonly used (Hess et al., 2005). This aspect should be investigated in future work. The results show that the MLM approach should be preferred to the MNL one to model exit choice. The interaction terms (i.e. the variables built by multiplying a dummy variable representing gender by each environmental and social variable, see Section 5) are not statistically different from zero. Therefore, our data do not prove any difference between male and female.

A behavioural analysis is performed considering the averages of the parameter distributions. In general, the probability of choosing an exit decreases when the number of evacuees close to it (NCE) increases, that is, the decision-makers perceive a large number of evacuees using an exit as an impedance. In other words, respondents show *crowd avoidance behaviour* with the evacuees near the two exits. The same tendency can be seen in the interaction with evacuees near the decision-maker (NCDM). However, the distribution of NCDM implies that the *crowd avoidance behaviour* is sometimes replaced by the *herding behaviour*. In fact, in Fig. 6, it can be seen that the probability is high when the parameter associated with NCDM is positive. This indicates that there are respondents for whom the fact that many other evacuees head towards one of the two exits is an incentive to select the same exit. This could be explained by the fear to be negatively judged by other evacuees by choosing the 'wrong' alternative (normative social influence (Lovreglio et al., 2016), and/or by the attitude to consider other people's decisions as a proof of the correctness of a choice (social proof theory) (Lovreglio et al., 2014, 2016; Cialdini, 1993). It is also evident from the figure that NCE and NCDM have different distributions, that is, there is a difference in the way respondents perceive these two factors, related to the same factor, the presence of other evacuees. This could be explained by the proxemics approach, which argues that the closer other people are to a decision-maker, the more the decision-maker is affected by them (Hall, 1966). This phenomenon has been observed also in virtual reality environments (Bailenson et al., 2003, 2008).

The exit distance is generally perceived as disutility as normally expected. However, the random distribution of the distance is characterised by a very large dispersion, that is, several respondents choose the furthest exit (Fig. 6). This could be because the two exits are not too far away from each other and therefore some participants did not consider distance as an important factor as most other did.

Considering the averages of the random parameters in Fig. 6, it is possible to argue that overall the respondents perceive the flow through the exits as a utility because higher flow rates allow faster evacuation, whereas the presence of smoke has a negative impact on the choice, as it could harm the decision-maker. Then, the presence of emergency lights increases, on average, the probability of choosing an exit, because it improves the functional affordance of the exit (Ronchi et al., 2015; Lovreglio et al., 2015). However, Fig. 6 shows that the parameters associated to FL, SM and EX have very large dispersions, as indicated by the standard deviations in Table 4. As a consequence, there is high probability that, for a specific decision-maker, the concerned parameters assume a sign different from that of the average decision-maker. The high dispersion can be due to a combination of the high level of heterogeneity in the preferences (Ortuzar and Willumsen, 2011).

Finally, the constant included in the utility function of the right-hand-side exit is not statistically significant ($p = 0.07 > 0.05$). This shows that the right-hand-side exit is not chosen systematically more than the left-hand-side one under the social/physical conditions described in Table 3.

6.1. Sensitivity analysis

A sensitivity analysis is conducted to show how the probability of choosing an exit is influenced by the observed factors and their interaction. A total of 300 β_z vector draws are randomly generated and the probabilities of choosing the two exits are estimated using Eq. (5). The effect of the number of evacuees near the two exits (NCE) and near the decision-maker (NCDM) is shown in Fig. 7, whereas the effect of flow (FL), smoke (SM), emergency lights (EL) and distances (DIST) is presented in Fig. 8.

In Fig. 7a and b, NCE and NCDM range between 0 and 50, whereas the other variables are the same for both exits. In the scenarios in which the variables are nil for one door and equal to 50 for the other, the probability of selecting the left-hand-side door is very close to 1 and 0 in Fig. 7a, whereas in Fig. 7b the probability surface has the maximum and minimum equal to 0.85 and 0.12, respectively. Comparing the two charts, it can be seen that NCE affects (negatively) the probability of choosing an exit more than NCDM. In other words, a decision-maker is more willing to choose the less congested exit when the other evacuees are closer to the exit (NCE) than when they are close to them (NCDM).

In Fig. 8, the number of evacuees close to the right-hand-side exit (NCE_R) is fixed to 25 while the evacuees close to the left-hand-side exit (NCE_L) vary between 0 and 50. All the other variables are the same for the two exits. In Fig. 8a and b, the flow and distance of the right-hand-side exit are 0.5 persons/s and 10 m, respectively.

Fig. 8a and b show the existence of situations in which flow and distance are not determinant in the exit choice, because the social factors are predominant. These situations occur when the curves in the figures tend to overlap. For instance, Fig. 8a

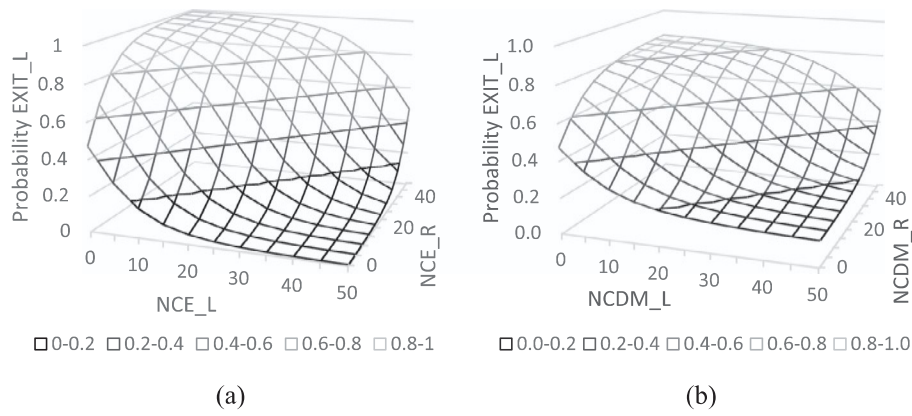


Fig. 7. Sensitivity analysis for (a) NCE; (b) NCDM.

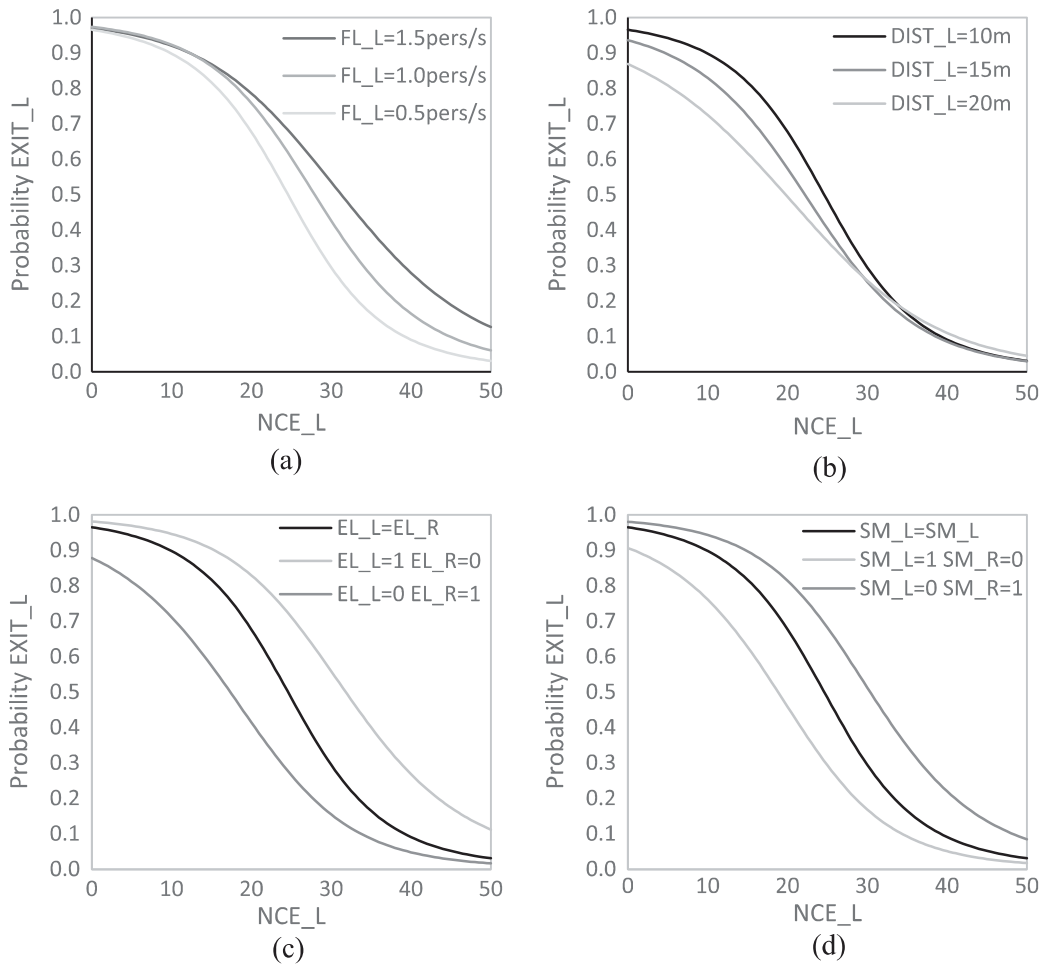


Fig. 8. Sensitivity analysis for (a) FL; (b) DIST; (c) EL; (d) SM.

shows that for low values of NCE_L ($NCE_L < 10$), the flow does not affect the choice. In these conditions, the left-hand-side exit is almost free and definitely freer than the right-hand-side one ($NCE_R = 25$) and therefore the decision-maker may reckon that he/she can escape quicker by using it, even though the capacity of the exit is low. It has to be noted that, for low values of NCE, the flow is difficult to evaluate for the decision-maker, and so he/she may assume that the flow rate is

Table 5

Comparison between the proposed model and the existing ones.

	Model specification						Survey features		Rsquared ^a
	NCE	FL	NCDM	SM	EL	DIST	Sample size	Video	
Duives and Mahmassani (2012)	Yes (−0.087) ^b	No (−)	No (−)	No (−)	No (−)	Yes (−0.055)	117	No	0.089
Lovreglio et al. (2014)	Yes (−0.134)	No (−)	Yes (−0.076)	No (−)	No (−)	Yes (−0.088)	191	Yes ^c	0.126
Haghani et al. (2014)	Yes (−0.089)	Yes (1.615)	No (−)	No (−)	No (−)	Yes (−0.082)	53	No	0.125
Proposed model	Yes (−0.116)	Yes (0.609)	Yes (−0.077)	Yes (−0.762)	Yes (0.856)	Yes (−0.053)	1503	Yes	0.208

^a Referring to the models fitted with the dataset collected in our survey.^b Mean of the parameter distribution.^c The degree on realism is definitely lower than the one of the videos used in this study.

the same for the two exits, leaving the number of evacuees as the only decision variable. In Fig. 8b, it can be seen that for high values of NCE_L (NCE_L > 30) the distance from the left-hand-side exit does not affect the choice probability, as this exit is so crowded that the decision-maker tends to avoid it anyway.

When the number of close evacuees is the same for the two exits as in Fig. 8c (NPE_L = NPE_R = 25), the probability of choosing the left-hand-side exit depends on the presence of the emergency lights. The probability varies from 0.26 when there is no light on the left-hand-side exit, but there is one on the right-hand-side one (EL_L = 0 and EL_R = 1), to 0.70 in the opposite case (EL_L = 1 and EL_R = 0). Fig. 8c suggests also that the decision-maker can neglect the information provided by the emergency light. In fact, in the situation with emergency light on the right-hand-side exit only, one would expect that the decision-makers avoid the exit on the left-hand side. Instead, the plot shows that the left-hand-side exit has high probability of being chosen for medium–small values of NCE_L. This could be explained by the informational social influence (Nilsson and Johansson, 2009), which predicts that the presence of other evacuees close to an exit indicates that that exit is an available alternative. Finally, Fig. 8d shows how the presence of smoke can affect the choice considering different numbers of evacuees near the exits. Having smoke close to an exit is less important than the choice of other evacuees. In fact, when there is smoke near the left-hand-side exit and the other exit is clear (SM_L = 1 and SM_R = 0), the decision-makers prefer the former alternative if it is relatively uncongested (NCE_L approaching zero).

6.2. Comparison with existing models

In Table 5, our model is compared to the existing ones. All the models have been fitted using our data set – that is definitely larger than all the others – for a fair comparison. It can be seen that the fit of our model in terms of adjusted R² indicator (which includes a penalty for each parameter included in the model) is much better than the others, proving the need for considering all the environmental and social factors together. The table shows also the mean values of the parameters (the parameters are proven random in all the cases) in each model specification. It can be seen that there is a remarkable difference between the parameter of FL (the exit flow rate) in the specification proposed by Haghani et al. (2014) and in our model. The under-specification of the former model may lead to wrong design choices whenever evacuation scenarios include smoke and emergency lighting. In fact, a design based on results by Haghani et al. (2014) may overestimate the possibility of inducing evacuees to select an exit by making it larger. However, the model specification proposed by Haghani et al. (2014) may still be correct in evacuations, which do not involve smoke and emergency lighting.

7. Conclusion

An MLM is proposed to predict exit choice in emergency at local level. The effect of different environmental and social factors is investigated. The model is estimated using the data collected through an online state preference survey designed through ED.

Compared to the existing literature, our study has the advantage to investigate the effect of more factors simultaneously expanding the current understanding of local exit choice in emergencies. The findings show that the presence of smoke, distance of the exit and the number of evacuees near the exit or close to the decision-maker, but moving towards the exit have a negative impact on the probability of an exit to be chosen. On the contrary, emergency lighting and flow of evacuees through the exit have a positive effect.

We explore the ‘intrinsic’ behavioural uncertainty and the behavioural uncertainty linked to ‘perceptions and preferences’ using MLM approach. In the estimated MLM, all the parameters are randomly distributed, which proves the existence of perceptions and preferences behavioural uncertainty. Methodologically, a strength of our study is the use of non-immersive virtual reality to represent choice scenarios. Through virtual reality, respondents have a substantially more realistic perception of the context to evaluate. This increases the validity of the survey. We asked participants to rate the

realism of the representation of the choice context. Previous studies did not include such an evaluation; therefore, a direct comparison is not possible.

The realism of our scenarios was rated 3 out of 5 on average, that is, the choice context was sufficiently realistic to respondents. However, we suggest that future studies should try to further enhance the realism of choice contexts. We note that, in the field of exit choice in emergency, when decision-makers are faced with unusual scenarios, the validity of results from simulated environments should be carefully assessed by means of comparison with naturalistic contexts. However, this is difficult because of the scarcity of data of real-world emergencies and the ethical implications of asking people to participate in experiments, which may harm their safety.

Our sample, made up of 1503 respondents from different parts of the world, is much larger than those used so far to investigate the local exit choice process. Our sample is rather homogeneous as to gender (72% are male) and age (71% are under 30 years of age) of participants, but it is definitely more heterogeneous than previous studies in terms of nationalities. Because of this heterogeneity, our results are not affected by the cultural attributes of the respondents linked to the nationality.

Our findings could have several applications. Once integrated with other models simulating strategic (i.e. pre-evacuation model) and operational (i.e. local movement) choices, our tactical model could improve the safety design and management of new and existing buildings (Bryant and Giachritsis, 2014; Galea, 2014). In fact, the model provides more accurate predictions of local exit choice, an aspect that has been previously simplified or overlooked in most evacuation models. We note that some possibly relevant factors, for example, number and position of the exits, are not taken into account in our model and their impact should be investigated in further studies.

Another important implementation of the proposed model concerns crowd management during evacuations from transportation systems. Recent studies have proved that it is possible to improve evacuation time and safety by using Intelligent Active Dynamic Signage System (IADSS) instead of using 'classic' passive signage system. The effectiveness of the system has been tested at the Sant Cugat Station in Barcelona, using volunteers. The experiments prove the improvement of wayfinding decisions in complex structures (see GETAWAY European Project for more details (Bryant and Giachritsis, 2014; Galea et al., 2015)). The aim of this system is to direct evacuees in real time by an iterative procedure having four steps: (1) detecting the *status quo* of evacuations (e.g. evacuees' position and movement direction and fire conditions); (2) predicting the outcomes due to the *status quo* conditions (e.g. total evacuation time and number of casualties); (3) verifying whether the evacuation process can be improved by directing the evacuees towards different exits/paths and (4) in case it could be improved, re-directing evacuees using dynamic signs. This approach crucially relies on evacuation modelling (steps 2 and 3) and an enhanced tactical model (e.g. global and local exit choice models) can improve the evacuation outcome, as IADSS systems are designed to support both the global and local evacuees' route choices using dynamic signs.

IADSS includes dissuasive emergency signage (Olander, 2015), a new generation of evacuation lighting not included in this study. Therefore, future work is necessary to investigate the effect of the dissuasive emergency signage, as our results show that the dynamic signs information may not be effective in determining exit choice (Section 6). Future studies should also investigate the effect of different smoke intensities (we only consider the presence or absence of smoke), which can affect the visibility of the exit and its affordance.

Our results may be biased by the demographic characteristics of the sample, in particular by the fact that most of the respondents are under 30 years of age and European. However, we have found no significant difference between the choices of female and male respondents. This result differs from the previous findings showing that the attitude to follow other people's choice depends on the gender (Lovreglio et al., 2014). The present study considers a broader set of explaining factors, which capture the effect previously attributed to gender.

This study was not designed with the purpose to recruit a specific population target but with the purpose to collect data from more than lower bound of the ED to investigate the behavioural uncertainty in local exit choice. However, analysing the sample demographics it is possible to characterise the 'target population' a posteriori and so to identify the best areas of application of the model. Our model is particularly suitable for any evacuation from an enclosed environment, in which the evacuees are young and coming from several nationalities, like evacuations from universities and underground stations in multinational cities at peak times (Lin and Lo, 2008). The methodology and the survey developed in this study can be used in future studies to investigate the behaviour of specific population targets defined a priori.

Finally, in the research field, the parameters estimated in this study can be used as a starting point for future SP studies based on ED and using more advanced techniques such as immersive virtual reality.

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Appendix A

The levels of the variables and the hypothetical scenarios of the pilot survey are shown in Tables A1 and A2, respectively.

Table A1

Levels for each variable included in the pilot survey.

Variable	Description	Levels
NCE _i ^a (pers)	Number of evacuees Close to the Exits	0 5 10 20
FL _i (pers/s)	Flow of evacuees through the exits	0.6 1.2 1.5
NCDM (pers)	Number of evacuees Close to the Decision-Maker	0 5 10
SM _i	Smoke near the exits	0 1
EL _i	Evacuation Lights above the exits	0 1
DIST (m)	Distance of the decision-maker from the exits	10 12 14 16
NEAR_E	Dummy variable equal to 0 if the decision-maker is closer to the right-hand exit, 1 otherwise	0 1
DIR	Dummy variable equal to 0 if the agents near the decision-maker move towards the right-hand exit, 1 otherwise	0 1

^a The values refer to the end of the videos.**Table A2**

Pilot scenarios.

Scenario	NCE _L	NCE _R	FL _L	FL _R	SM _L	SM _R	EL _L	EL _R	NCDM	DIR	DIST	NEAR_E
1	20	0	0.9	0.9	1	1	0	0	5	1	1	20
2	10	20	0.6	1.2	0	1	0	0	5	1	0	10
3	10	0	1.5	1.2	0	1	0	1	0	0	1	10
4	5	20	1.5	0.6	0	1	1	0	0	1	0	5
5	20	5	0.6	1.2	0	0	1	1	0	0	1	20
6	10	10	0.6	0.6	1	1	1	1	5	1	1	10
7	5	10	1.2	0.9	0	0	0	1	5	1	0	5
8	5	0	1.2	1.5	0	0	1	0	5	1	1	5
9	0	20	0.9	1.5	1	0	0	1	0	0	0	0
10	0	5	1.5	0.6	1	0	1	1	5	1	0	0
11	20	10	1.2	0.9	1	0	0	0	0	1	1	20
12	0	5	0.9	1.5	1	1	1	0	0	0	0	0

Appendix B. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.tra.2016.06.018>.

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